

WHY THE UNDEFORMED $h = 2$ MINIMUM-MODULUS PLOT RAMPS AND THEN PLATEAUS: A FIXED-PRECISION ROOT-FINDING ARTIFACT

CLAUDE

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ABSTRACT. The plot of minimum-modulus eigenvalues μ_n for the undeformed sequence $h_0 = 1$, $h_n = 2$ ($n \geq 1$) rises linearly to $n \approx 125$ and then settles into a noisy horizontal band at $\mu \approx 36$ out to $n = 500$. We show that the rising part is the true behaviour — μ_n grows without bound like κn with $\kappa = 0.278465\dots$ — and that the plateau is entirely an artifact of computing the roots of the characteristic polynomial in fixed (here 100-bit) precision. The location of the plateau is set by the working precision, not by the mathematics, and we give a one-line test that distinguishes this numerical wall from a genuine bounded saturation (such as the deformed cases, whose plateaus are real). Nonstandard terms are defined where they first appear (the Szegő curve in §2; conditioning, the Wilkinson phenomenon and the companion matrix in §3; Newton’s method, catastrophic cancellation, the incomplete gamma function and the Aberth–Ehrlich method in §5).

1. THE PHENOMENON

The characteristic polynomial is formed *exactly* over $\mathbb{Q}$ by Theorem 2.3 of the oscillating-spectra manuscript,

$$\chi_n(x) = \sum_{r=0}^n \binom{n}{r} r! h_r (-1)^r x^{n-r} \quad (h_0 = 1, h_r = 2 \text{ for } r \geq 1),$$

so the coefficients carry no error. The computation then calls a numerical root-finder at 100 bits of precision (Sage’s `chi.roots(ring=ComplexField(100))`) and records $\mu_n = \min_{\chi_n(x)=0} |x|$. The resulting curve ramps with slope ≈ 0.27 to about $(125, 36)$ and is flat and noisy thereafter.

The contrast between a faithful high-precision computation and ordinary double precision already exposes what is happening:

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n	double (53-bit)	true (high precision)	κn
40	12.05	12.05	11.14
60	17.56	17.66	16.71
80	18.29	23.25	22.28
100	17.92	28.84	27.85
120	18.22	34.43	33.42

The high-precision column keeps climbing along κn ; the double-precision column stops climbing near $n \approx 65$ and freezes at ≈ 18 . The 100-bit run in the notebook is the same effect with the wall moved further out (to $n \approx 125$, height ≈ 36). See Figure 1.

2. THE RAMP IS REAL: A THEOREM OF SZEGŐ

Because $h_0 = 1$ and $h_r = 2$, the partial-fraction identity gives $\chi_n(x) = 2(-1)^n n! E_n(-x) - x^n$, where

$$E_n(z) = \sum_{k=0}^n \frac{z^k}{k!}$$

is the n -th partial sum of the exponential series (the Taylor polynomial of e^z truncated after the z^n term). The subtracted x^n is negligible near the smallest root — its size relative to the first term is $O((\kappa e)^n / \sqrt{n})$ with $\kappa e = 0.757 < 1$ — so the smallest-modulus root of χ_n is, to exponentially good approximation, the smallest-modulus zero of $E_n(-x)$.

The location of those zeros is classical.

Theorem 2.1 (Szegő, 1924; see [10, 4, 14]). *Let E_n be the n -th partial sum of e^z . As $n \rightarrow \infty$ the rescaled zeros z/n of $E_n(z)$ accumulate on the Szegő curve*

$$\mathcal{S} = \left\{ z \in \mathbb{C} : |z e^{1-z}| = 1, |z| \leq 1 \right\}.$$

The Szegő curve $\mathcal{S}$ is a teardrop-shaped loop through $z = 1$, symmetric about the real axis and contained in the closed unit disc; it is simply the set of limit points of the (rescaled) roots of these truncated exponentials. Writing $z = r e^{i\theta}$, the defining equation $|z e^{1-z}| = 1$ becomes $r e^{1-r \cos \theta} = 1$, i.e. $r \cos \theta = 1 + \ln r$. The point of $\mathcal{S}$ nearest the origin occurs at $\theta = \pi$ (the negative real direction), where $\cos \theta = -1$ forces

$$r + \ln r = -1, \quad \text{whose unique solution in } (0, 1) \text{ is} \quad \kappa = 0.278464542761 \dots$$

Hence the smallest zero of $E_n(z)$ sits near $z = -\kappa n$, the smallest zero of $E_n(-x)$ near $x = +\kappa n$, and therefore

$$\mu_n \sim \kappa n, \quad \kappa = 0.278465 \dots$$

The slope you see in the ramp is exactly κ , and μ_n is *unbounded*: at $n = 500$ the true value is near 139, not 36.

3. THE PLATEAU IS SPURIOUS: CONDITIONING AND THE PRECISION WALL

What a numerical root-finder is up against. Two facts about the roots of χ_n combine badly. First, the constant term is $\chi_n(0) = (-1)^n n! h_n = \pm 2n!$, so the product of all n roots has magnitude $2n!$ and the *typical* root modulus is $(2n!)^{1/n} \approx n/e$: the roots all lie at distance proportional to n from the origin. Second, by Theorem 2.1 they are not scattered there but packed onto the single curve $n\mathcal{S}$, so they form a tight *cluster*. Clustered roots are the textbook source of extreme sensitivity.

Condition number and Wilkinson’s phenomenon. The *condition number* of a problem measures how much its answer can move when its input is perturbed; a problem is *ill-conditioned* when a tiny relative change in the input forces a large change in the output, so that finite-precision arithmetic cannot deliver an accurate answer no matter how good the algorithm. For a simple root ζ of a polynomial p , a perturbation δp of the coefficients moves the root by

$$\delta\zeta \approx -\frac{\delta p(\zeta)}{p'(\zeta)}.$$

When the other roots crowd around ζ , the derivative $p'(\zeta) = \prod_j (\zeta - \zeta_j)$ is minuscule, so $\delta\zeta$ is enormous: the root is hypersensitive. This is exactly the effect J. H. Wilkinson made famous with his “perfidious polynomial,” whose roots $1, 2, \dots, 20$ move by huge amounts under changes in the coefficients of size 2^{-23} [12, 13]; modern textbook treatments are [11, Lect. 12–14] and [7, Ch. 25]. For our cluster on $n\mathcal{S}$ the root condition number grows *exponentially* in n .

The wall, and why it moves with precision. A floating-point computation carried out with b bits of mantissa introduces relative perturbations of size 2^{-b} . The computed smallest root remains accurate only while $2^{-b} \times (\text{condition number}) \lesssim 1$, i.e. while

$$b \gtrsim \log_2(\text{condition number}) \propto n.$$

Once n exceeds this precision-determined threshold, the solver can no longer resolve the smallest root; the reported minimum modulus stops tracking κn and freezes near the last value it could resolve, jittering with round-off. Empirically the threshold is about *one bit of precision per unit of n* (more precisely, ≈ 0.8): 53-bit (double) arithmetic walls out near $n \approx 65$ at height ≈ 18 , while 100-bit arithmetic walls out near $n \approx 125$ at height ≈ 36 . The ratio of knee positions, $125/65 = 1.92$, matches the ratio of precisions, $100/53 = 1.89$, and in each case the plateau height equals κ times the knee position. A genuine mathematical saturation cannot depend on the number of bits the solver happens to use; this one does. (The single-root method of §5 lowers this constant from ≈ 0.8 to ≈ 0.6 bit per unit n , but does not remove the linear growth in the precision required.)

(Two common back-ends both exhibit the wall. `numpy.roots` builds the *companion matrix*—the matrix whose characteristic polynomial is χ_n —and computes its eigenvalues in double precision; Sage’s `.roots(ring=ComplexField(b))` runs a b -bit simultaneous root iteration. The wall is a property of the *conditioning*, so it appears regardless of which is used; only its location depends on b .)

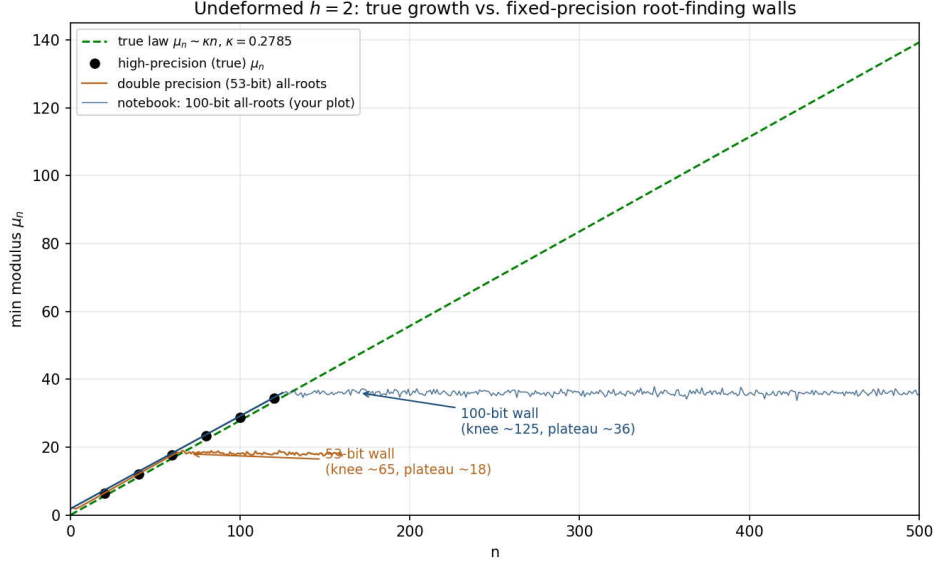


FIGURE 1. The true law $\mu_n \sim \kappa n$ (green dashed; black dots are high-precision values) versus fixed-precision all-roots computations. Double precision (orange) tracks the truth to $n \approx 65$, then walls out near 18. The notebook’s 100-bit run (blue) tracks further, to $n \approx 125$, then walls out near 36 — the shape in the uploaded plot. The plateau height equals κ times the knee position; raising the precision slides the knee to the right along the green line.

4. A DIAGNOSTIC, AND THE DEFORMED CONTRAST

This ramp-then-noisy-plateau shape is the generic signature of *compute-all-roots, then take the minimum modulus* applied to these determinant polynomials, because (as above) their roots always sit clustered at distance $\propto n$ from the origin. The shape recurs across the calculations precisely for that structural reason.

To decide in any given plot whether a plateau is real or a wall, there is a one-line test:

Recompute a few of the plateau points with the working precision doubled. If the plateau rises, it is the numerical wall; if it stays put, the saturation is genuine.

The deformed cases are the genuine kind. There the smallest eigenvalue is truly $O(1)$ — pinned exactly at $x = c$ for odd n (the truncated sinh vanishes at the origin), and near $c \pm i\pi/2$ for even n — so those bounded plateaus are precision-independent and trustworthy. The undeformed plateau is the artifact kind, because the true minimum modulus is unbounded.

5. RECOVERING THE TRUE VALUES

Two routes avoid the wall. The second is the safe default; the first is genuinely cheaper but subtler than an earlier draft of this note suggested, so we describe it carefully and correct that draft's overstatement.

(i) Find only the smallest root. The minimum modulus is a *single* root, and Theorem 2.1 predicts its location in advance, near $x_0 = \kappa n$. One can therefore target it directly with *Newton's method* — the iteration $x \mapsto x - f(x)/f'(x)$ for solving $f(x) = 0$, which once x is near a simple root roughly doubles the number of correct digits per step, so a handful of steps from a good start suffices [9, Ch. 9]. The natural choice is $f(x) = E_n(-x)$, whose derivative is $f'(x) = -E_{n-1}(-x)$, so each step reads $x \mapsto x + E_n(-x)/E_{n-1}(-x)$. Computing one root this way sidesteps the conditioning of the whole cluster (§3) that wrecks the all-roots approach, and it is far faster: one root from scalar evaluations instead of an $n \times n$ eigenproblem. Two cautions are essential, and both were missing from the earlier draft.

The start must be complex. The partial sum E_n has no real zeros when n is even and exactly one when n is odd — an elementary induction from $E'_n = E_{n-1}$ (for odd n , $E_{n-1} > 0$ everywhere, so E_n is increasing; for even n , the lone critical point is a minimum with positive value $x_0^n/n!$); see [14]. Hence for even n the smallest-modulus zero of $E_n(-x)$ is a complex-conjugate pair sitting just off the real axis near $x = \kappa n$. A real starting value cannot converge to it; take $x_0 = \kappa n + i$.

It is not precision-free. The bottleneck is not locating the root but evaluating f . Near its zero, $E_n(-x) = \sum_{k=0}^n (-x)^k/k!$ is an alternating sum whose largest term has size $\approx e^{\kappa n}$ while the total is near 0; forming it in fixed precision loses on the order of $0.4n$ – $0.6n$ bits to *catastrophic cancellation* (the loss of significant digits when nearly equal quantities are subtracted, so that the small true result is swamped by rounding error in the large terms [7, Ch. 1–2], the standard short exposition being [6]). Consequently the naive Newton iteration, while cheaper than the all-roots wall (≈ 0.8 bit per unit n), still needs precision growing with n :

working precision	$ x $ accurate up to about
15 digits ($\approx$ double)	$n \approx 80$
30 digits	$n \approx 200$
50 digits	$n \approx 250\text{--}300$

i.e. roughly $\text{dps} \gtrsim n/5$ (≈ 0.6 bit per unit n). The gain over the all-roots method is therefore a constant factor in precision and a large factor in speed, *not* the removal of the precision dependence claimed earlier.

The cancellation can in principle be removed by evaluating E_n in closed form through the (*upper*) *incomplete gamma function* $\Gamma(s, w) = \int_w^\infty t^{s-1} e^{-t} dt$, via the exact identity (valid for integer n ; [5, Eq. 8.4.8])

$$E_n(w) = \frac{e^w \Gamma(n+1, w)}{n!},$$

so that the zeros of $E_n(-x)$ are exactly the zeros of $\Gamma(n+1, -x)$, which good libraries evaluate by stable continued fractions and asymptotics rather than by cancelling sums [5, §8.9, §8.11]. In practice this is not a clean substitute: at low working precision the library evaluation of $\Gamma(n+1, -x)$ for large n is itself unreliable, and a bare secant or Newton step on it tends to wander to a larger-modulus zero unless carefully safeguarded. We record the identity for completeness but do not rely on it.

A tested implementation of route (i) (`mpmath`; the Sage equivalent uses `ComplexField(bits)` with $\text{bits} \approx 0.7n + 50$ and the same Newton step) is:

```
import mpmath as mp
def mu_smallest(n, dps=None):
    if dps is None:
        dps = n//5 + 20          # grow precision with n
    mp.mp.dps = dps
    kappa = mp.mpf('0.27846454276')
    def En(w, m):
        # E_m(w) = sum w^k/k!
        s = mp.mpf(0); t = mp.mpf(1)
        for k in range(m + 1):
            if k: t = t * w / k
            s += t
        return s
    x = mp.mpc(kappa * n, 1)      # complex start (even n!)
    for _ in range(100):
        # Newton, f = E_n(-x)
        w = -x
        xn = x - En(w, n) / (-En(w, n - 1))
        if abs(xn - x) < mp.mpf(10)**(-dps + 6):
            x = xn; break
    x = xn
    return abs(x)
```

At adequate `dps` this reproduces the high-precision column of §1; for instance $n = 500$ returns $|x| \approx 140.5 = \kappa \cdot 500$ at `dps` ≈ 120 .

(ii) **If all roots are wanted, scale precision with n or use an adaptive solver.** Because the threshold is about one bit per unit n , an honest all-roots computation to $n = 500$ needs on the order of 500 bits. Better, use a solver designed for exactly this difficulty. The *Aberth–Ehrlich method* refines all n root estimates *simultaneously*: each estimate takes a Newton step from which the pull of the other current estimates is subtracted, so the estimates repel one another and do not collapse onto each other in a cluster [1, 2]. Its multiprecision implementation MPSOLVE [3] raises the internal precision automatically until each root meets a requested accuracy, and is the standard tool for ill-conditioned or clustered roots. (The closely related *Durand–Kerner*, a.k.a. *Weierstrass*, method — used by `mpmath.polyroots` — is another simultaneous iteration, but without adaptive precision it simply fails to converge once the cluster outruns the working precision, which is the failure mode encountered while preparing this note.)

In the present problem neither route is strictly necessary: since $\mu_n/n \rightarrow \kappa = 0.278465\dots$ in closed form, the quantity is known without computation, and the plateau in the figure should be read as the point at which the 100-bit solver lost the smallest root, not as a feature of the sequence.

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